EXPERIMENT NO. 2

Common Values Between Two Arrays

Programming Language: Java

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Aim

To write a Java program that counts how many elements of each array also exist in the other array.

Question

You are given two integer arrays nums1 and nums2 of sizes n and m, respectively. Calculate the following values:

answer1: the number of indices i such that nums1[i] exists in nums2.

answer2: the number of indices i such that nums2[i] exists in nums1.

Return [answer1, answer2].

Example 1

Input: nums1 = [2,3,2], nums2 = [1,2]

Output: [2,1]

Example 2

Input: nums1 = [4,3,2,3,1], nums2 = [2,2,5,2,3,6]

Output: [3,4]

Explanation: The elements at indices 1, 2, and 3 in nums1 exist in nums2 as well. So answer1 is 3. The elements at indices 0, 1, 3, and 4 in nums2 exist in nums1. So answer2 is 4.

Algorithm

Step 1: Start.

Step 2: Store the arrays nums1 and nums2, compute their sizes n and m, and set answer1 = 0 and answer2 = 0.

Step 3: For each element nums1[i], search nums2 linearly. If a match is found, increment answer1 and stop searching for that element.

Step 4: For each element nums2[i], search nums1 linearly. If a match is found, increment answer2 and stop searching for that element.

Step 5: Print [answer1, answer2].

Step 6: Stop.

Program (Java)

```
public class Main {
```

```
    static int countCommon(int[] a, int[] b) {
        int count = 0;
        for (int i = 0; i < a.length; i++) {
            for (int j = 0; j < b.length; j++) {
                if (a[i] == b[j]) {
                    count++;
                    break;      // found, stop searching for a[i]
                }
            }
        }
        return count;
    }

    public static void main(String[] args) {
        int[] a1 = {2, 3, 2},       b1 = {1, 2};
        int[] a2 = {4, 3, 2, 3, 1}, b2 = {2, 2, 5, 2, 3, 6};

        System.out.println("Output: [" + countCommon(a1, b1) + ", " +
countCommon(b1, a1) + "]");
        System.out.println("Output: [" + countCommon(a2, b2) + ", " +
countCommon(b2, a2) + "]");
    }
}
```

Output

```
Output: [2, 1]
Output: [3, 4]

=== Code Execution Successful ===
```

Complexity Analysis

Time Complexity: O(n * m) in the worst case, O(n + m) in the best case.
Space Complexity: O(1)

Result

Thus the program was executed successfully and the output was verified.